



Blacktown Boys' High School

2019

HSC Trial Examination

Mathematics Extension 2

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided for this paper
- All diagrams are not drawn to scale
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

**Total marks:
100****Section I – 10 marks** (pages 3 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8 – 15)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Assessor: X. Chirgwin

Student Name: _____

Teacher Name: _____

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the 2019 Higher School Certificate Examination.

Office Use Only

Question	Mark
Q1	/1
Q2	/1
Q3	/1
Q4	/1
Q5	/1
Q6	/1
Q7	/1
Q8	/1
Q9	/1
Q10	/1
Q11 a)	/1
Q11 b)	/2
Q11 c)	/4
Q11 d)	/3
Q11 e)	/5
Q12 a)	/4
Q12 b)	/3
Q12 c)	/8
Q13 a)	/5
Q13 b)	/2
Q13 c)	/8
Q14 a)	/3
Q14 b)	/3
Q14 c)	/4
Q14 d)	/5
Q15 a)	/5
Q15 b)	/6
Q15 c)	/4
Q16 a)	/5
Q16 b)	/5
Q16 c)	/5
Total	/100

Section I

10 marks

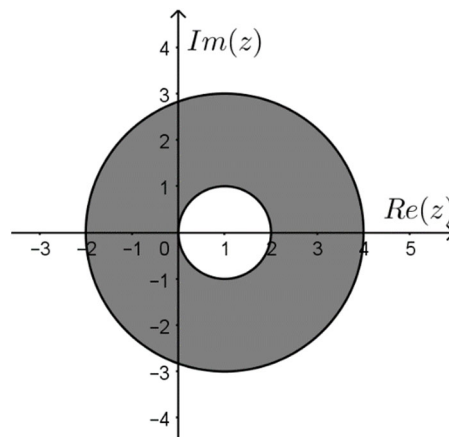
Attempt Questions 1–10

Use the multiple choice answer sheet provided on page 17 for Questions 1–10.

1 Let $z = 7 + 10i$. What is the value of $\bar{iz}$?

- A. $-10 + 7i$
- B. $-10 - 7i$
- C. $10 + 7i$
- D. $10 - 7i$

2 Consider the Argand diagram below.



Which inequality could define the shaded area?

- A. $1 \leq |z + 1| \leq 3$
- B. $2 \leq |z + 1| \leq 4$
- C. $1 \leq |z - 1| \leq 3$
- D. $2 \leq |z - 1| \leq 4$

3 What are the equations of the asymptotes of the hyperbola $9y^2 - 25x^2 = 225$?

A. $y = \pm \frac{25}{9}x$

B. $y = \pm \frac{9}{25}x$

C. $y = \pm \frac{5}{3}x$

D. $y = \pm \frac{3}{5}x$

4 The sum of the eccentricities of two different conics is $\frac{3}{2}$.

Which pair of the conics could this be?

A. Circle and ellipse

B. Circle and parabola

C. Parabola and hyperbola

D. Hyperbola and circle

5 Which of the following parametric equations represent $3(x - 5)^2 + 4y^2 = 12$

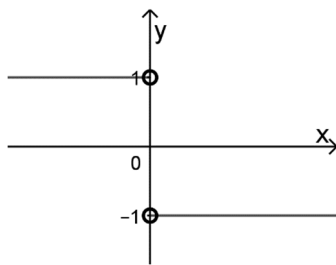
A. $x = 2 \cos \theta + 5, y = \sqrt{3} \sin \theta$

B. $x = 2 \cos \theta - 5, y = \sqrt{3} \sin \theta$

C. $x = \sqrt{3} \cos \theta + 5, y = 2 \sin \theta$

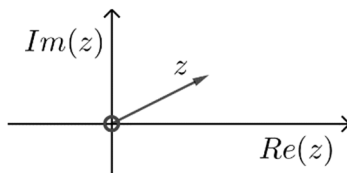
D. $x = 3 \cos \theta + 5, y = 4 \sin \theta$

- 6 Given that $f(x) = |x|$ and $g(x) = x$.

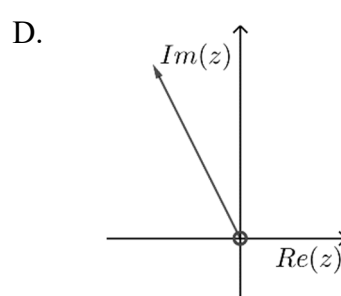
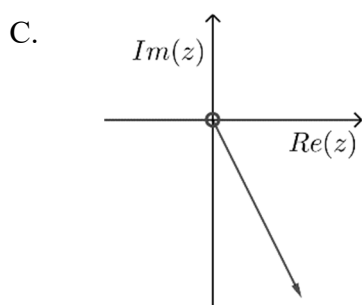
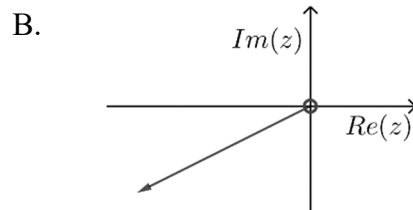
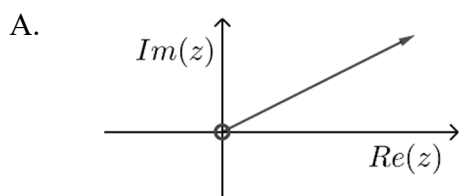


Which of the following statement is true for the graph shown above?

- A. $y = \frac{f(x)}{g(x)}$
- B. $y = -\frac{f(x)}{g(x)}$
- C. $y = f(x) \times g(x)$
- D. $y = -f(x) \times g(x)$
- 7 The Argand diagram below shows the complex number z .



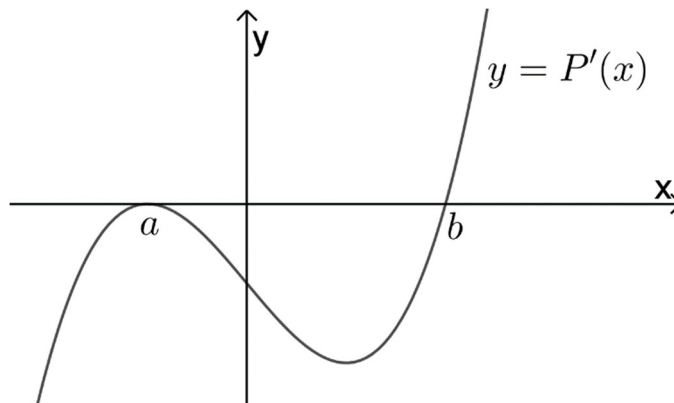
Which Argand diagram best represents $\frac{2z}{i}$?



- 8 There are 9 identical boxes on a table. One person comes to the table and takes three boxes. 2 more people do the same as the first person. How many ways can this happen?

A. 362880
 B. 60480
 C. 3360
 D. 1680

- 9 The following diagram shows the graph $y = P'(x)$, the derivative of a polynomial $P(x)$. $P'(x)$ has roots at $x = a$ and $x = b$ where $a < 0$ and $b > 0$.



If c is a root of $P(x)$, where $c > b$. Which of the following could be $P(x)$?

A. $P(x) = (x - a)^3(x - c)$
 B. $P(x) = (x - a)^3(x + c)$
 C. $P(x) = (x + a)^3(x - c)$
 D. $P(x) = (x + a)^3(x + c)$

10 Given that a is a constant, which of the following is equivalent to $\int_0^a x(a-x)^{99} dx$?

A. $\frac{a}{100}$

B. $\frac{a}{101}$

C. $\frac{a^{100}}{10000}$

D. $\frac{a^{101}}{10100}$

End of Section I

Section II**90 Marks****Attempt Questions 11–16**

Answer each question on a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- a) Simplify i^{2019} . 1
- b) Sketch $\frac{x^2}{9} + \frac{y^2}{25} = 1$, indicating the coordinates of its foci and equations of the directrices. 2
- c) i) Express $\sqrt{2} - \sqrt{6}i$ in modulus-argument form. 2
ii) Hence evaluate $(\sqrt{2} - \sqrt{6}i)^8$ in the form $x + iy$. 2
- d) Evaluate $\int_2^4 \sqrt{16 - x^2} dx$ 3
- e) i) Find the domain and range of $f(x) = x \sin^{-1} x$. 2
ii) Determine whether $f(x)$ is even, odd, or neither. 1
iii) Sketch $y = f(x)$, showing clearly the end values. 2

End of Questions 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

a) Given that $P(x) = x^5 + 6x^4 + 16x^3 + 32x^2 + 48x + 32$.

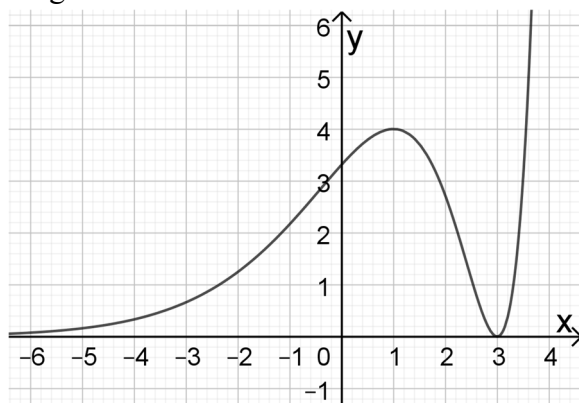
i) Show that $x = -2$ is a root of $P(x)$ of multiplicity three. 2

ii) Hence, or otherwise, factorise $P(x)$ completely into linear factors. 2

b) Use the substitution $t = \tan \frac{x}{2}$, or otherwise, show that 3

$$\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} = \frac{\sqrt{3}\pi}{9}$$

c) The function $f(x) = (x - 3)^2 e^{x-1}$ has stationary points at $x = 1$ and $x = 3$ as shown in the diagram below.



Draw separate half-page sketches of the following graphs. In each case, label any asymptotes and the coordinates of any turning points.

i) $y = f(|x|)$ 2

ii) $y = \frac{1}{f(x)}$ 2

iii) $y = [f(x)]^2$ 2

iv) $y^2 = f(x)$ 2

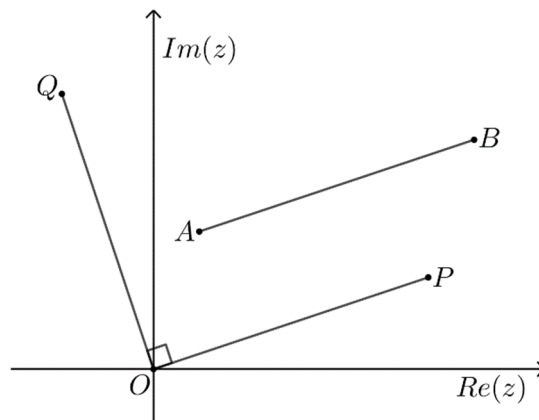
End of Questions 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

a) The equation $x^3 - 5x + 25 = 0$ has roots α, β and γ . Find:

- i) the polynomial with roots $-\alpha, -\beta, -\gamma$. 1
- ii) the polynomial with roots $\alpha^2, \beta^2, \gamma^2$. 2
- iii) the value of $\alpha^3 + \beta^3 + \gamma^3$. 2

b) In the Argand diagram below, intervals AB , OP and OQ are equal in length, OP is parallel to AB and $\angle POQ = \frac{\pi}{2}$. If A and B represent the complex number $1 + 3i$ and $7 + 5i$ respectively, find the complex number which is represented by Q . 2

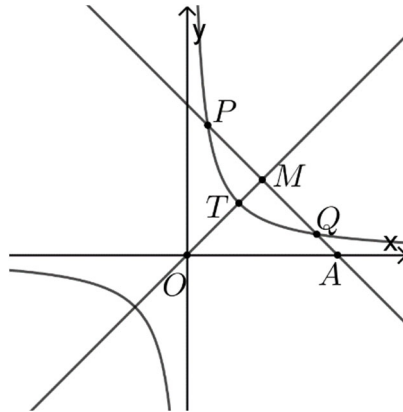


Question 13 continues on page 11

Question 13 (continued)

- c) In the diagram $P(cp, cp^{-1})$ and $Q(cq, cq^{-1})$ where $0 < p < q$, are points on the hyperbola $xy = c^2$.

M is the midpoint of PQ and the line PQ cuts the x -axis at A . OM cuts the hyperbola at T .

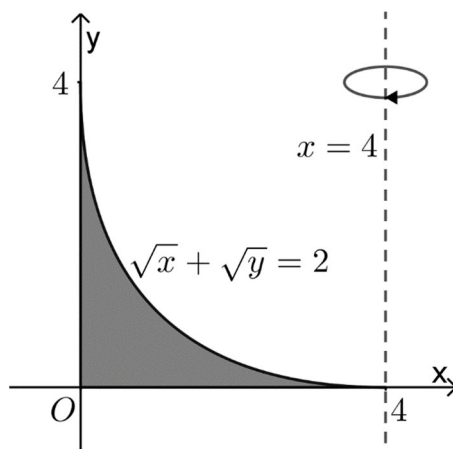


- | | | |
|------|---|----------|
| i) | Show that gradient of OM is $-1 \times$ gradient of MA . | 2 |
| ii) | Hence, or otherwise, show that $OM = MA$. | 1 |
| iii) | Show that the tangent to the hyperbola at T is parallel to the chord PQ . | 2 |
| iv) | Find the coordinates of R and S where the tangent to the hyperbola at T cuts the x and y axes respectively. | 2 |
| v) | Prove that the area of the triangle ORS is a constant. | 1 |

End of Questions 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- a) The region bounded by the curve $\sqrt{x} + \sqrt{y} = 2$ and the x -axis between $x = 0$ and $x = 4$ is rotated about the line $x = 4$.



- i) Use the method of cylindrical shells to show that the volume V of the solid formed is given by: 2
- $$V = 2\pi \int_0^4 (16 - 16\sqrt{x} + 4x\sqrt{x} - x^2) dx$$
- ii) Hence find the exact volume V . 1
- b) A solid has ellipse $16x^2 + 49y^2 = 784$ as its base. If each section perpendicular to the major axis is an equilateral triangle, find the volume of this solid. 3
- c) A eleven-member Wellbeing Committee consists of five students, four teachers, and two parents. The committee meets around a circular table.
- i) How many different arrangements of the eleven members around the table are possible if the students sit together as a group and so do the teachers, but no teacher sits next to a student? 2
- ii) One student and one parent are related. Given that all arrangements in part i) are equally likely, what is the probability that these two members sit next to each other? 2

Question 14 continues on page 13

Question 14 (continued)

- d) i) By considering $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2$, where $x > 0$, prove that the sum of a positive real number and its reciprocal is never less than 2, and is only equal to 2 when $x = 1$. **2**
- ii) Hence, or otherwise, find the smallest value of $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right)$ where a and b are positive real numbers. For what values of a and b does this minimum value occur? **3**

End of Questions 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- a) A particle of mass 3 kg moves under the action of a force F so that the velocity is given by $v = 10\sqrt{4 - x^2} \text{ ms}^{-1}$.
- Find the force F in terms of the displacement x . 2
 - If the particle is initially 2 metres to the right of the origin. Find the displacement of the particle at any time t . 3
- b) A particle of mass m falls from rest under gravitational acceleration $g \text{ ms}^{-2}$ and air resistance proportional to the speed $v \text{ ms}^{-1}$. If the constant of proportionality is k .
- Explain why the acceleration is $g - kv$ for this particle. 1
 - Find the velocity of the particle at time t . 2
 - Find the terminal velocity. 1
 - Express the distance the particle has fallen at time t in terms of t . 2
- c) Given that $I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$, where n is a positive integer.
- Show that $I_{2n+1} = \frac{1}{2}e - nI_{2n-1}$. 3
 - Hence, or otherwise, evaluate $\int_0^1 x^3 e^{x^2} dx$. 1

End of Questions 15

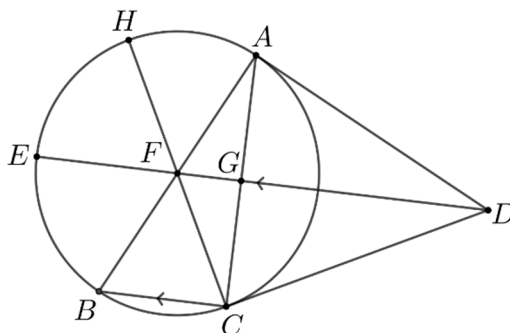
Question 16 (15 marks) Use a SEPARATE writing booklet.

- a) A sequence $u_1, u_2, u_3, u_4, \dots$ satisfies the relationship $u_n = u_{n-1} + u_{n-2}$ for $n \geq 3$.

i) Show that $u_1 u_2 + u_2 u_3 = u_3^2 - u_1^2$. 1

ii) For $n \geq 1$, use mathematical induction to show that 4
 $u_1 u_2 + u_2 u_3 + u_3 u_4 + \dots + u_{2n-1} u_{2n} + u_{2n} u_{2n+1} = u_{2n+1}^2 - u_1^2$

- b) The triangle ABC is inscribed in a circle. From an external point D , tangents are drawn to the circle, touching it at A and C . The chord ED is drawn parallel to BC , meeting AB at F and AC at G . The line CF is produced to meet the circle at H .



i) Prove that $AFCD$ is a cyclic quadrilateral. 2

ii) Prove that $HF = AF$. 3

- c) i) If $8x + 4y = \pi$, show that 2

$$\tan y = \frac{1 - 2 \tan x - \tan^2 x}{1 + 2 \tan x - \tan^2 x}$$

ii) Hence deduce that $\tan \frac{\pi}{8}$ is a root of the equation 3

$$t^2 + 2t - 1 = 0, \text{ and find the exact value of } \tan \frac{\pi}{8}.$$

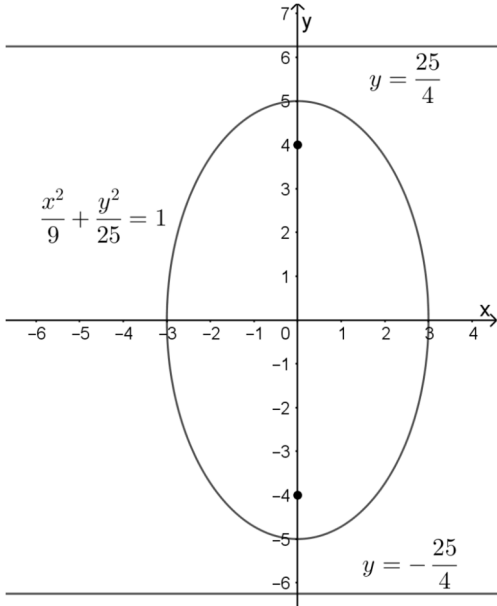
End of Paper

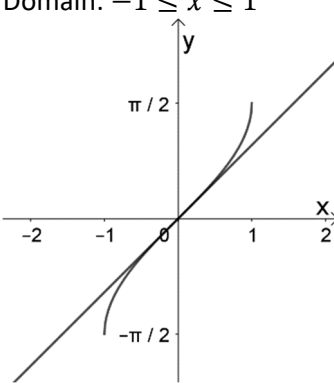
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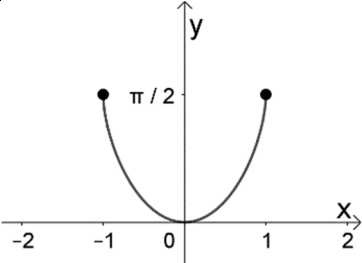
Section 1

1	B $z = 7 + 10i$ $iz = -10 + 7i$ $i\bar{z} = -10 - 7i$	1 Mark
2	C Circle centre at $(1, 0)$, inner circle radius 1 and outer circle radius 3 $1 \leq z - 1 \leq 3$	1 Mark
3	C $9y^2 - 25x^2 = 225$ $\frac{y^2}{25} - \frac{x^2}{9} = 1$ Asymptotes: $y = \pm \frac{5}{3}x$	1 Mark
4	D Circle has $e = 0$ Ellipse has $e < 1$ Parabola has $e = 1$ Hyperbola has $e > 1$ $e = \frac{3}{2}$ Circle and ellipse $e < 1$ Circle and parabola $e = 1$ Parabola and hyperbola $e > 2$ Hyperbola and circle $e > 1$ $\therefore$ Hyperbola and circle could be the pair	1 Mark
5	A $x = 2 \cos \theta + 5, y = \sqrt{3} \sin \theta$ Sub into $3(x - 5)^2 + 4y^2 = 12$ $LHS = 3(2 \cos \theta + 5 - 5)^2 + 4(\sqrt{3} \sin \theta)^2$ $LHS = 12 \cos^2 \theta + 12 \sin^2 \theta$ $LHS = 12$ $LHS = RHS$	1 Mark
6	B $y = -\frac{ x }{x}$ $y = -\frac{f(x)}{g(x)}$	1 Mark
7	C $2z$ is doubling the length of z and then dividing by i indicates rotating it clockwise by $\frac{\pi}{2}$, so the resulting vector should be in the 4 th quadrant.	1 Mark

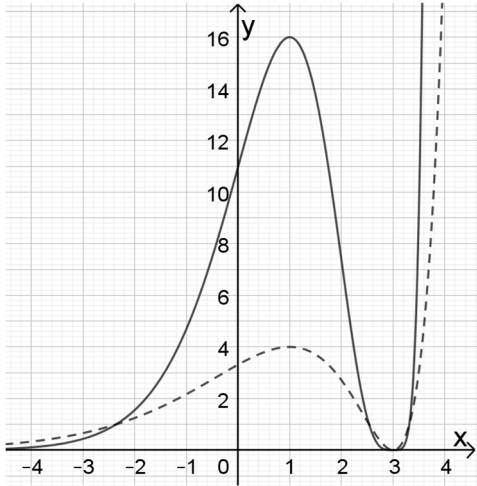
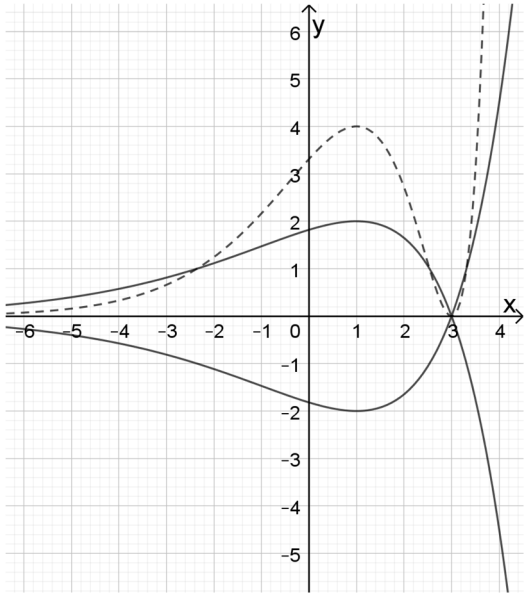
8	D First person can select 3 out of 9 boxes = 9C_3 ways Second person can select 3 out of 6 remaining boxes = 6C_3 ways Last person can select the remaining 3 presents in 1 way = 3C_3 way Number of ways = ${}^9C_3 \times {}^6C_3 \times 1 = 1680$	1 Mark
9	A $P'(x)$ has a double root at $x = a$, so $P(x)$ could have a triple root at $x = a$ and a turning point at $x = b$, and given that $x = c$ is a root of $P(x)$ to the right of b. $\therefore P(x) = (x - a)^3(x - c)$ is the possible solution	1 Mark
10	D $\int_0^a x(a - x)^{99} dx$ $= \int_0^a (a - x)x^{99} dx$ $= \int_0^a (ax^{99} - x^{100}) dx$ $= \left[\frac{ax^{100}}{100} - \frac{x^{101}}{101} \right]_0^a$ $= \left(\frac{a \times a^{100}}{100} - \frac{a^{101}}{101} \right) - 0$ $= \left(\frac{a^{101}}{100} - \frac{a^{101}}{101} \right)$ $= \left(\frac{101a^{101}}{10100} - \frac{100a^{101}}{10100} \right)$ $= \frac{a^{101}}{10100}$	1 Mark

Section 2		
Q11 a)	i^{2019} $= i^{2016} \times i^3$ $= (i^4)^{504} \times i^2 \times i$ $= 1 \times (-1) \times i$ $= -i$	1 Mark Correct solution
Q11 b)	$\frac{x^2}{9} + \frac{y^2}{25} = 1$ $a = 3, b = 5$ $a^2 = b^2(1 - e^2)$ $9 = 25(1 - e^2)$ $\frac{9}{25} = 1 - e^2$ $e^2 = 1 - \frac{9}{25}$ $e^2 = \frac{16}{25}$ $e = \frac{4}{5}$ Foci: $(0, \pm 4)$ Directrices: $y = \pm \frac{25}{4}$ 	2 Marks Correct graph with all key features shown clearly 1 Mark Obtains the correct value of eccentricity
Q11 c) i)	$ \sqrt{2} - \sqrt{6}i = \sqrt{(\sqrt{2})^2 + (\sqrt{6})^2}$ $ \sqrt{2} - \sqrt{6}i = 2\sqrt{2}$ $\arg(\sqrt{2} - \sqrt{6}i) = \tan^{-1}\left(\frac{-\sqrt{6}}{\sqrt{2}}\right)$ $\arg(\sqrt{2} - \sqrt{6}i) = -\frac{\pi}{3}$ $\sqrt{2} - \sqrt{6}i = 2\sqrt{2}\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$ $\sqrt{2} - \sqrt{6}i = 2\sqrt{2}\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$	2 Marks Correct solution 1 Mark Correct modulus or argument
Q11 c) ii)	$(\sqrt{2} - \sqrt{6}i)^8 = \left[2\sqrt{2}\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)\right]^8$ Using De Moivre's theorem $(\sqrt{2} - \sqrt{6}i)^8 = (2\sqrt{2})^8 \left(\cos\frac{8\pi}{3} - i\sin\frac{8\pi}{3}\right)$ $(\sqrt{2} - \sqrt{6}i)^8 = (2\sqrt{2})^8 \left(-\frac{1}{2} - i \times \frac{\sqrt{3}}{2}\right)$ $(\sqrt{2} - \sqrt{6}i)^8 = 2^{12} \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$ $(\sqrt{2} - \sqrt{6}i)^8 = -2^{11}(1 + \sqrt{3}i) \text{ or } -2048(1 + \sqrt{3}i)$	2 Marks Correct solution 1 Mark Applies De Moivre's theorem correctly

Q11 d)	$I = \int_2^4 \sqrt{16 - x^2} dx$ Let $x = 4 \cos \theta$ $dx = -4 \sin \theta d\theta$ $x = 4, \quad \theta = 0$ $x = 2, \quad \theta = \frac{\pi}{3}$ $I = \int_{\frac{\pi}{3}}^0 \sqrt{16 - 16 \cos^2 \theta} \times -4 \sin \theta d\theta$ $I = 16 \int_0^{\frac{\pi}{3}} \sqrt{1 - \cos^2 \theta} \times \sin \theta d\theta$ $I = 16 \int_0^{\frac{\pi}{3}} \sin^2 \theta d\theta$ $I = \frac{16}{2} \int_0^{\frac{\pi}{3}} (1 - \cos 2\theta) d\theta$ $I = 8 \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{3}}$ $I = 8 \left[\frac{\pi}{3} - \frac{1}{2} \sin \left(2 \times \frac{\pi}{3} \right) - 0 \right]$ $I = 8 \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right]$ $I = \frac{8\pi}{3} - 2\sqrt{3}$	3 Marks Correct solution 2 Marks Finds a correct primitive 1 Mark Uses correct trigonometric substitution
Q11 e) i)	$f(x) = x \sin^{-1} x$ Domain: $-1 \leq x \leq 1$  For $-1 \leq x < 0$, $y = x$ is negative, $y = \sin^{-1} x$ is also negative, so $x \sin^{-1} x$ must be positive. For $0 < x \leq 1$, $y = x$ is positive, $y = \sin^{-1} x$ is also positive, so $x \sin^{-1} x$ must be positive. For $x = 0$, $y = x$ is zero, $y = \sin^{-1} x$ is also zero, so $x \sin^{-1} x$ must be zero. Range: $0 \leq y \leq \frac{\pi}{2}$	2 Marks Correct solution 1 Mark Correct domain or range
Q11 e) ii)	$f(x) = x \sin^{-1} x$ $f(-x) = (-x) \times \sin^{-1}(-x)$ $f(-x) = (-x) \times -\sin^{-1} x$ $f(-x) = x \sin^{-1} x$ $f(x) = f(-x)$ $\therefore f(x)$ is an even function	1 Mark Correct solution

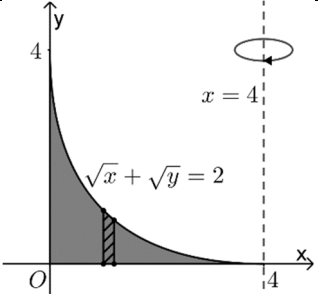
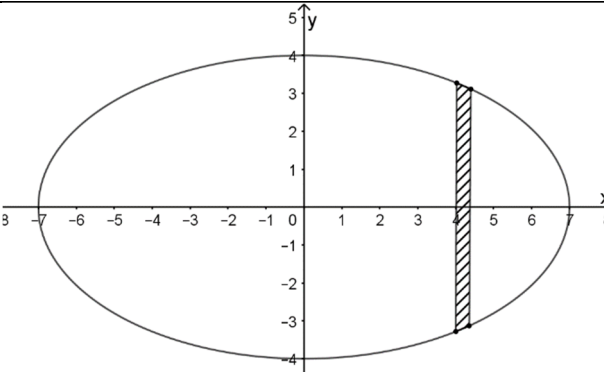
Q11 e) iii)		2 Marks Correct sketch 1 Mark Sketch shows $f(x)$ is even with correct domain and range
Q12 a) i)	$P(x) = x^5 + 6x^4 + 16x^3 + 32x^2 + 48x + 32$ $P'(x) = 5x^4 + 24x^3 + 48x^2 + 64x + 48$ $P''(x) = 20x^3 + 72x^2 + 96x + 64$ $P''(-2) = 20(-2)^3 + 72(-2)^2 + 96(-2) + 64$ $P''(-2) = 0$ $P'(-2) = 5(-2)^4 + 24(-2)^3 + 48(-2)^2 + 64(-2) + 48$ $P'(-2) = 0$ $P(-2) = (-2)^5 + 6(-2)^4 + 16(-2)^3 + 32(-2)^2 + 48(-2) + 32$ $P(-2) = 0$ $\therefore x = -2$ is a root of multiplicity 3.	2 Marks Correct solution 1 Mark Show that $P''(-2) = 0$
Q12 a) ii)	Roots of $P(x)$ are $\alpha, \beta, -2, -2, -2$ Sum of roots: $\alpha + \beta + (-2) + (-2) + (-2) = -6$ $\alpha + \beta = 0$ Product of roots: $\alpha\beta \times (-2)^3 = -32$ $\alpha\beta = 4$ Let α and β are the solutions to $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ $x^2 + 4 = 0$ $x = \pm 2i$ $\therefore P(x) = (x + 2)^3(x - 2i)(x + 2i)$	2 Marks Correct solution 1 Mark Finds product and sum of remaining two roots
Q12 b)	$t = \tan \frac{x}{2}$ $\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$ $\frac{dt}{dx} = \frac{1}{2} \left(1 + \tan^2 \frac{x}{2} \right)$ $\frac{dt}{dx} = \frac{1}{2} (1 + t^2)$ $dx = \frac{2dt}{1 + t^2}$ $x = \frac{\pi}{2}, \quad t = 1$ $x = 0, \quad t = 0$ $\sin x = \frac{2t}{1 + t^2}$	3 Marks Correct solution 2 Marks Finds a correct primitive 1 Mark Attempts to obtain an integral in terms of t

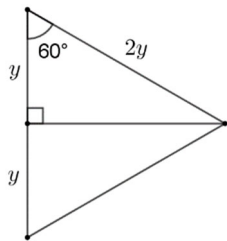
	$\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x}$ $= \int_0^1 \frac{1}{2 + \frac{2t}{1+t^2}} \times \frac{2dt}{1+t^2}$ $= \int_0^1 \frac{1+t^2}{2+2t^2+2t} \times \frac{2dt}{1+t^2}$ $= \int_0^1 \frac{dt}{t^2+t+1}$ $= \int_0^1 \frac{dt}{t^2+t+\frac{1}{4}+\frac{3}{4}}$ $= \int_0^1 \frac{dt}{\left(t+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$ $= \left[\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) \right]_0^1$ $= \left[\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) \right]_0^1$ $= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2 \times 1 + 1}{\sqrt{3}} \right) - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2 \times 0 + 1}{\sqrt{3}} \right)$ $= \frac{2}{\sqrt{3}} \tan^{-1} \sqrt{3} - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$ $= \frac{2}{\sqrt{3}} \times \frac{\pi}{3} - \frac{2}{\sqrt{3}} \times \frac{\pi}{6}$ $= \frac{3\sqrt{3}}{\sqrt{3}\pi}$ $= \frac{\pi}{9}$	
Q12 c) i)	$y = f(x)$	2 Marks Correct sketch 1 Mark Recognises reflection along the y-axis
Q12 c) ii)	$y = \frac{1}{f(x)}$	2 Marks Correct sketch 1 Mark Recognises and labels the asymptote

Q12 c) iii)	$y = [f(x)]^2$ 	2 Marks Correct sketch 1 Mark Correct key points shown
Q12 c) iv)	$y^2 = f(x)$ 	2 Marks Correct sketch 1 Mark Correct key points shown
Q13 a) i)	$x^3 - 5x + 25 = 0$ Let $x = -\alpha$ $\alpha = -x$ $(-x)^3 - 5(-x) + 25 = 0$ $-x^3 + 5x + 25 = 0$ $x^3 - 5x - 25 = 0$	1 Mark Correct solution
Q13 a) ii)	$x^3 - 5x + 25 = 0$ Let $x = \alpha^2$ $\alpha = \pm\sqrt{x}$ $(\pm\sqrt{x})^3 - 5(\pm\sqrt{x}) + 25 = 0$ $\pm\sqrt{x}(x - 5) = -25$ $x(x - 5)^2 = 625$ $x(x^2 - 10x + 25) = 625$ $x^3 - 10x^2 + 25x - 625 = 0$	2 Marks Correct solution 1 Mark Recognises the relationships of the roots and makes the correct substitution

Q13 a) iii)	$x^3 - 5x + 25 = 0$ $\alpha^3 - 5\alpha + 25 = 0 \quad (1)$ $\beta^3 - 5\beta + 25 = 0 \quad (2)$ $\gamma^3 - 5\gamma + 25 = 0 \quad (3)$ $(1) + (2) + (3)$ $\alpha^3 + \beta^3 + \gamma^3 - 5(\alpha + \beta + \gamma) + 25 \times 3 = 0$ $\alpha^3 + \beta^3 + \gamma^3 = 5(\alpha + \beta + \gamma) - 75$ $\alpha^3 + \beta^3 + \gamma^3 = -75$	2 Marks Correct solution 1 Mark Recognises the relationships of the roots and makes the correct substitution
Q13 b)	$\overrightarrow{AB} = (7 + 5i) - (1 + 3i)$ $\overrightarrow{AB} = 6 + 2i$ $\overrightarrow{OP} = 6 + 2i$ $\overrightarrow{OQ} = i\overrightarrow{OP}$ $\overrightarrow{OQ} = i(6 + 2i)$ $\overrightarrow{OQ} = -2 + 6i$	2 Marks Correct solution 1 Mark Finds the complex number that represent P
Q13 c) i)	$M\left(\frac{cp + cq}{2}, \frac{\frac{c}{p} + \frac{c}{q}}{2}\right)$ $M\left(\frac{c}{2}(p + q), \frac{c}{2}\left(\frac{p + q}{pq}\right)\right)$ $m_{OM} = \frac{\frac{c}{2}\left(\frac{p + q}{pq}\right) - 0}{\frac{c}{2}(p + q) - 0}$ $m_{OM} = \frac{1}{pq}$ $m_{MA} = m_{PQ}$ $m_{MA} = \frac{\frac{c}{p} - \frac{c}{q}}{\frac{cp - cq}{q - p}}$ $m_{MA} = \frac{\frac{pq}{q - p}}{p - q}$ $m_{MA} = -\frac{1}{pq}$ $-1 \times m_{MA} = \frac{1}{pq}$ $\therefore m_{OM} = -1 \times m_{MA}$	2 Marks Correct solution 1 Mark Finds the gradient of OM or MA
Q13 c) ii)	$\tan \angle MOA = m_{OM}$ $\tan \angle MOA = \frac{1}{pq}$ $\tan \angle MAx = m_{MA}$ $\tan \angle MAx = -\frac{1}{pq}$ $\angle MAx = \pi - \angle MOA$ $\therefore \angle MAO = \angle MOA$ $\therefore OM = MA \text{ (equal sides opposite equal angles in a triangle)}$	1 Mark Correct solution

Q13 c) iii)	Equation of OM $y = \frac{1}{pq}x$ T is the intersection point of OM and hyperbola $xy = c^2$ $x \times \frac{1}{pq}x = c^2$ $x^2 = c^2 pq$ $x = c\sqrt{pq}$ $y = c^2 x^{-1}$ $\frac{dy}{dx} = -c^2 x^{-2}$ At $x = c\sqrt{pq}$ Gradient of tangent at T $m_T = \frac{-c^2}{(c\sqrt{pq})^2}$ $m_T = -\frac{1}{pq}$ $m_T = m_{PQ}$ $\therefore$ tangent at T is parallel to chord PQ.	2 Marks Correct solution 1 Mark Find the x value of T
Q13 c) iv)	At T $x = c\sqrt{pq}$ $y = \frac{c\sqrt{pq}}{pq}$ Equation of tangent at T $y - \frac{c\sqrt{pq}}{pq} = -\frac{1}{pq}(x - c\sqrt{pq})$ $pqy - c\sqrt{pq} = -x + c\sqrt{pq}$ $x + pqy - 2c\sqrt{pq} = 0$ At $R, y = 0$ $x = 2c\sqrt{pq}$ $R(2c\sqrt{pq}, 0)$ At $S, x = 0$ $pqy - 2c\sqrt{pq} = 0$ $y = \frac{2c\sqrt{pq}}{pq}$ $S\left(0, \frac{2c\sqrt{pq}}{pq}\right)$	2 Marks Correct solution 1 Mark Finds the equation of tangent at T
Q13 c) v)	$A = \frac{1}{2} \times OR \times OS$ $A = \frac{1}{2} \times 2c\sqrt{pq} \times \frac{2c\sqrt{pq}}{pq}$ $A = 2c^2$ $\therefore$ Area of triangle ORS is a constant since c is a constant.	1 Mark Correct solution

Q14 a) i)	 $A(x) = 2\pi(4 - x)y$ $\sqrt{y} = 2 - \sqrt{x}$ $y = (2 - \sqrt{x})^2$ $\Delta V = 2\pi(4 - x)y\Delta x$ $\Delta V = 2\pi(4 - x)(2 - \sqrt{x})^2 \Delta x$ $V = 2\pi \int_0^4 (4 - x)(2 - \sqrt{x})^2 dx$ $V = 2\pi \int_0^4 (4 - x)(4 - 4\sqrt{x} + x) dx$ $V = 2\pi \int_0^4 (16 - 16\sqrt{x} + 4x - 4x + 4x\sqrt{x} - x^2) dx$ $V = 2\pi \int_0^4 (16 - 16\sqrt{x} + 4x\sqrt{x} - x^2) dx$	2 Marks Correct solution 1 Mark Finds the correct area in terms of x
Q14 a) ii)	$V = 2\pi \int_0^4 (16 - 16\sqrt{x} + 4x\sqrt{x} - x^2) dx$ $V = 2\pi \left[16x - \frac{16x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{4x^{\frac{5}{2}}}{\frac{5}{2}} - \frac{x^3}{3} \right]_0^4$ $V = 2\pi \left[\left(16 \times 4 - \frac{16 \times 4^{\frac{3}{2}}}{\frac{3}{2}} + \frac{4 \times 4^{\frac{5}{2}}}{\frac{5}{2}} - \frac{4^3}{3} \right) - 0 \right]$ $V = 2\pi \left(\frac{128}{15} \right)$ $V = \frac{256\pi}{15} \text{ units}^3$	1 Mark Correct solution
Q14 b)	 $16x^2 + 49y^2 = 784$ $\frac{x^2}{49} + \frac{y^2}{16} = 1$	3 Marks Correct solution 2 Marks Makes significant progress in finding the volume 1 Mark Finds the correct cross sectional area



Cross-sectional area

$$A = \frac{1}{2} \times 2y \times 2y \times \sin 60^\circ$$

$$A = \sqrt{3}y^2$$

$$\Delta V = \sqrt{3}y^2 \Delta x$$

$$\Delta V = \sqrt{3} \times 16 \left(1 - \frac{x^2}{49} \right) \Delta x$$

$$V = 16\sqrt{3} \int_{-7}^7 \left(1 - \frac{x^2}{49} \right) dx$$

$$V = 16\sqrt{3} \left[x - \frac{x^3}{147} \right]_{-7}^7$$

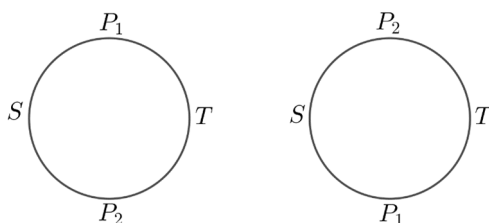
$$V = 16\sqrt{3} \left[\left(7 - \frac{7^3}{147} \right) - \left((-7) - \frac{(-7)^3}{147} \right) \right]$$

$$V = 16\sqrt{3} \times \frac{28}{3}$$

$$V = \frac{448\sqrt{3}}{3} \text{ units}^3$$

Q14 c) i)

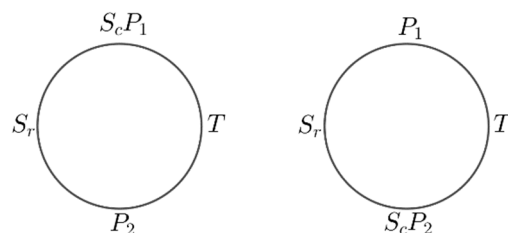
If S and T denote the sets 'students' and 'teachers' respectively, and P_1, P_2 are the parents, then all the possible arrangements are shown below.



There are $5!$ ways of arranging the students and $4!$ ways of arranging the teachers. So the total number of arrangements of the 11 members is $2 \times 5! \times 4! = 5760$.

Q14 c) ii)

Let S_c be the related student. They could be related to either P_1 or P_2 . Let S_r denote the remaining set of students and T be the set of teachers. Then the possible arrangements are shown below.



There are $4!$ Ways of arranging the students S_r and $4!$ Ways of arranging the teachers T . Thus the number of arrangements of the 11 members is $2 \times 4! \times 4! = 1152$

Hence the probability of this pair of people sit together is:

$$\frac{1152}{5760} = \frac{1}{5}$$

2 Marks

Correct solution

1 Mark

Recognises the arrangement of students and teachers

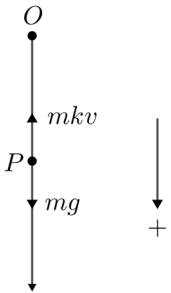
2 Marks

Correct solution

1 Mark

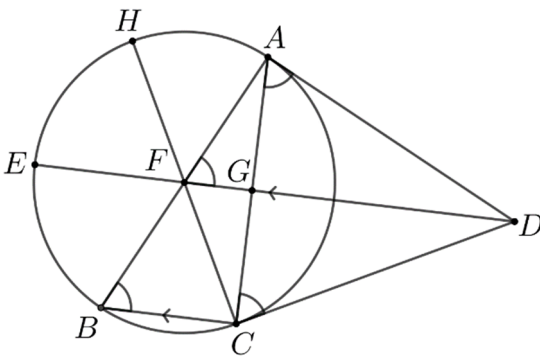
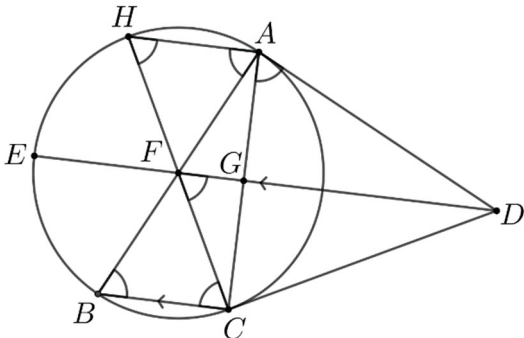
Finds the correct number of arrangements

Q14 d) i)	$\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = x - 2 + \frac{1}{x}$ $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 \geq 0$ $x - 2 + \frac{1}{x} \geq 0$ $x + \frac{1}{x} \geq 2$ Hence the sum of a positive number and its reciprocal is never less than 2. If $x = 1$, then $x + \frac{1}{x} = 2$	2 Marks Correct solution 1 Mark Deduce $x + \frac{1}{x} \geq 2$
Q14 d) ii)	$(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{a}{b} + \frac{b}{a} + 2$ Let $A = \frac{a}{b}$ $\frac{1}{A} = \frac{b}{a}$ Using Part i) $A + \frac{1}{A} \geq 2$ $\frac{a}{b} + \frac{b}{a} \geq 2$ $\frac{a}{b} + \frac{b}{a} + 2 \geq 4$ $\therefore (a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$ Hence the minimum value is 4 and this minimum value occurs when $a = b$.	3 Marks Correct solution 2 Marks Show that the minimum value is 4 1 Mark Shows $\frac{a}{b} + \frac{b}{a} \geq 2$
Q15 a) i)	$v = 10\sqrt{4 - x^2}$ $v = 10(4 - x^2)^{\frac{1}{2}}$ $\ddot{x} = v \frac{dv}{dx} = 10(4 - x^2)^{\frac{1}{2}} \times 10 \times \frac{1}{2} \times -2x \times (4 - x^2)^{-\frac{1}{2}}$ $\ddot{x} = -100x$ Or $\frac{1}{2}v^2 = \frac{1}{2} \times \left(10(4 - x^2)^{\frac{1}{2}}\right)^2$ $\frac{1}{2}v^2 = 50(4 - x^2)$ $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 50 \times -2x$ $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -100x$ $F = m\ddot{x}$ $F = -300x$	2 Marks Correct solution 1 Mark Expresses acceleration in terms of displacement

Q15 a) ii)	$v = \frac{dx}{dt} = 10(4 - x^2)^{\frac{1}{2}}$ $\frac{dt}{dx} = \frac{1}{10(4 - x^2)^{\frac{1}{2}}}$ $t = \int \frac{1}{10(4 - x^2)^{\frac{1}{2}}} dx$ $t = \frac{1}{10} \int \frac{1}{\sqrt{4 - x^2}} dx$ $t = \frac{1}{10} \sin^{-1} \frac{x}{2} + C$ When $x = 2, t = 0$ $0 = \frac{1}{10} \sin^{-1} \frac{2}{2} + C$ $C = -\frac{\pi}{20}$ $t = \frac{1}{10} \sin^{-1} \frac{x}{2} - \frac{\pi}{20}$ $t + \frac{\pi}{20} = \frac{1}{10} \sin^{-1} \frac{x}{2}$ $10t + \frac{\pi}{2} = \sin^{-1} \frac{x}{2}$ $\frac{x}{2} = \sin\left(10t + \frac{\pi}{2}\right)$ $\therefore x = 2 \sin\left(10t + \frac{\pi}{2}\right)$	3 Marks Correct solution 2 Marks Finds the correct value of C 1 Mark Expresses t as a primitive function in terms of x
Q15 b) i)	Take O as the point of release of the particle P. Take motion downwards as positive. There are two forces acting on P: its weight force acting downwards and its resistance of mkv acting upwards.  $F = mg - mkv$ $m\ddot{x} = mg - mkv$ $\ddot{x} = g - kv$	1 Mark Correct solution
Q15 b) ii)	$\frac{dv}{dt} = g - kv$ $\frac{dt}{dv} = \frac{1}{g - kv}$ $t = \int \frac{1}{g - kv} dv$ $t = -\frac{1}{k} \ln(g - kv) + C$	2 Marks Correct solution 1 Mark Find the primitive function of t in terms of v with correct constant value

	$t = 0, v = 0$ $0 = -\frac{1}{k} \ln g + C$ $C = \frac{1}{k} \ln g$ $t = -\frac{1}{k} \ln(g - kv) + \frac{1}{k} \ln g$ $t = \frac{1}{k} \ln \left(\frac{g}{g - kv} \right)$ $kt = \ln \left(\frac{g}{g - kv} \right)$ $e^{kt} = \frac{g}{g - kv}$ $\frac{g - kv}{g} = \frac{1}{e^{kt}}$ $g - kv = g e^{-kt}$ $-kv = g e^{-kt} - g$ $kv = g(1 - e^{-kt})$ $v = \frac{g}{k} (1 - e^{-kt})$	
Q15 b) iii)	$t \rightarrow \infty, e^{-kt} \rightarrow 0$ $v \rightarrow \frac{g}{k} (1 - 0)$ $v \rightarrow \frac{g}{k}$	1 Mark Correct solution
Q15 b) iv)	$\frac{dx}{dt} = \frac{g}{k} (1 - e^{-kt})$ $x = \frac{g}{k} \int (1 - e^{-kt}) dt$ $x = \frac{g}{k} \left[t + \frac{1}{k} e^{-kt} \right] + C$ $t = 0, x = 0$ $0 = \frac{g}{k} \left[0 + \frac{1}{k} e^0 \right] + C$ $C = -\frac{g}{k^2}$ $x = \frac{g}{k} \left[t + \frac{1}{k} e^{-kt} \right] - \frac{g}{k^2}$ $x = \frac{g}{k} \left(t + \frac{1}{k} e^{-kt} - \frac{1}{k} \right)$	2 Marks Correct solution 1 Mark Find the primitive function
Q15 c) i)	$I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$ $u = x^{2n} \qquad v' = x e^{x^2}$ $u' = 2n x^{2n-1} \qquad v = \frac{1}{2} e^{x^2}$	3 Marks Correct solution 2 Marks Applies integration by parts correctly

	$I_{2n+1} = \int_0^1 x^{2n} \times x e^{x^2} dx$ $I_{2n+1} = \left[x^{2n} \times \frac{1}{2} e^{x^2} \right]_0^1 - \int_0^1 2nx^{2n-1} \times \frac{1}{2} e^{x^2} dx$ $I_{2n+1} = \left[\left(1^{2n} \times \frac{1}{2} e^1 \right) - 0 \right] - n \int_0^1 x^{2n-1} e^{x^2} dx$ $I_{2n+1} = \frac{1}{2} e - n I_{2n-1}$	1 Mark Correctly identifies u and v' and finds corresponding u' and v
Q15 c) ii)	$I_3 = \int_0^1 x^3 e^{x^2} dx$ $I_3 = \frac{1}{2} e - 1 \times I_1$ $I_1 = \int_0^1 x e^{x^2} dx$ $I_1 = \left[\frac{1}{2} e^{x^2} \right]_0^1$ $I_1 = \left[\frac{1}{2} e^1 - \frac{1}{2} e^0 \right]_0^1$ $I_1 = \frac{1}{2} (e - 1)$ $I_3 = \frac{1}{2} e - 1 \times \frac{1}{2} (e - 1)$ $I_3 = \frac{1}{2}$	1 Marks Correct solution
Q16 a) i)	$u_n = u_{n-1} + u_{n-2}$ $u_3 = u_2 + u_1$ $u_2 = u_3 - u_1$ $u_1 u_2 + u_2 u_3$ $= u_2 (u_1 + u_3)$ $= (u_3 - u_1)(u_3 + u_1)$ $= u_3^2 - u_1^2$	1 Mark Correct solution
Q16 a) ii)	$u_1 u_2 + u_2 u_3 + u_3 u_4 + \dots + u_{2n-1} u_{2n} + u_{2n} u_{2n+1} = u_{2n+1}^2 - u_1^2$ 1. Prove statement is true for $n = 1$ $LHS = u_1 u_2 + u_2 u_3$ $RHS = u_{2 \times 1 + 1}^2 - u_1^2$ $RHS = u_3^2 - u_1^2$ $u_1 u_2 + u_2 u_3 = u_3^2 - u_1^2$ (proven in part i)) $LHS = RHS$ $\therefore$ statement is true for $n = 1$ 2. Assume statement is true for $n = k$ $u_1 u_2 + u_2 u_3 + u_3 u_4 + \dots + u_{2k-1} u_{2k} + u_{2k} u_{2k+1} = u_{2k+1}^2 - u_1^2$ 3. Prove statement is true for $n = k + 1$ i.e. $(u_1 u_2 + u_2 u_3 + u_3 u_4 + \dots + u_{2k-1} u_{2k} + u_{2k} u_{2k+1}) +$ $u_{2k+1} u_{2k+2} + u_{2k+2} u_{2k+3} = u_{2k+3}^2 - u_1^2$ $LHS = (u_1 u_2 + u_2 u_3 + u_3 u_4 + \dots + u_{2k-1} u_{2k} + u_{2k} u_{2k+1})$ $\quad + u_{2k+1} u_{2k+2} + u_{2k+2} u_{2k+3}$ $LHS = u_{2k+1}^2 - u_1^2 + u_{2k+2} (u_{2k+1} + u_{2k+3})$ (from step 2)	4 Marks Correct solution 3 Marks Makes significant progress in the body of proof 2 Marks Correct $k + 1$ statement 1 Mark Prove initial statement is true

	Since $u_n = u_{n-1} + u_{n-2}$ $u_{2k+3} = u_{2k+2} + u_{2k+1}$ $u_{2k+2} = u_{2k+3} - u_{2k+1}$ $LHS = u_{2k+1}^2 - u_1^2 + (u_{2k+3} - u_{2k+1})(u_{2k+3} + u_{2k+1})$ $LHS = u_{2k+1}^2 - u_1^2 + u_{2k+3}^2 - u_{2k+1}^2$ $LHS = u_{2k+3}^2 - u_1^2$ $LHS = RHS$ $\therefore$ statement is true by mathematical induction for all $n \geq 1$	
Q16 b) i)	 $\angle CAD = \angle ABC$ (Alternate segment theorem) $\angle ABC = \angle AFD$ (corresponding angles are equal, $ED \parallel BC$) $AD = CD$ (tangents from an external point are equal in length) $\angle CAD = \angle ACD$ (angles opposite equal sides of $\triangle ACD$ are equal) $\therefore \angle AFD = \angle ACD$ Since angles in the same segment are equal $\therefore AFCD$ is a cyclic quadrilateral	2 Marks Correct solution 1 Mark Makes significant progress
Q16 b) ii)	Join HA  Since $AFCD$ is a cyclic quadrilateral $\angle DAC = \angle DFC$ (angles in the same segment are equal) $\angle FCB = \angle DFC$ (alternate angles are equal, $ED \parallel BC$) Since $\angle FBC = \angle FCB$ (proven above that $\angle CAD = \angle ABC$) $\therefore \triangle FBC$ is an isosceles triangle $\angle HAB = \angle HCB$ (angles in the same segment) $\angle AHC = \angle ABC$ (angles in the same segment) $\therefore \triangle HFA \parallel \triangle CFB$ (equiangular) $\therefore \triangle HFA$ is also an isosceles triangle $\therefore HF = AF$ (equal sides of isosceles triangle)	3 Marks Correct solution 2 Marks Shows $\triangle HFA \parallel \triangle CFB$ 1 Marks Shows $\angle FBC = \angle FCB$

Q16 c) i)	$8x + 4y = \pi$ $y = \frac{\pi}{4} - 2x$ $\tan y = \tan\left(\frac{\pi}{4} - 2x\right)$ $\tan y = \frac{\tan \frac{\pi}{4} - \tan 2x}{1 + \tan \frac{\pi}{4} \tan 2x}$ $\tan y = \frac{1 - \tan 2x}{1 + \tan 2x}$ $\tan y = \frac{1 - \frac{2 \tan x}{1 - \tan^2 x}}{1 + \frac{2 \tan x}{1 - \tan^2 x}}$ $\tan y = \frac{1 - \tan^2 x - 2 \tan x}{1 - \tan^2 x + 2 \tan x}$ $\tan y = \frac{1 - \tan^2 x - 2 \tan x}{1 - \tan^2 x + 2 \tan x}$	2 Marks Correct solution 1 Mark Finds $\tan y = \frac{1 - \tan 2x}{1 + \tan 2x}$
Q16 c) ii)	If $y = 0$ $0 = \frac{\pi}{4} - 2x$ $x = \frac{\pi}{8}$ $\tan 0 = \frac{1 - \tan^2 \frac{\pi}{8} - 2 \tan \frac{\pi}{8}}{1 - \tan^2 \frac{\pi}{8} + 2 \tan \frac{\pi}{8}}$ $1 - \tan^2 \frac{\pi}{8} - 2 \tan \frac{\pi}{8} = 0$ Let $t = \tan \frac{\pi}{8}$ $1 - t^2 - 2t = 0$ $t^2 + 2t - 1 = 0$ $\therefore \tan \frac{\pi}{8}$ must a root of of $t^2 + 2t - 1 = 0$ $t = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times (-1)}}{2 \times 1}$ $t = \frac{-2 \pm \sqrt{8}}{2}$ $t = -1 \pm \sqrt{2}$ $\tan \frac{\pi}{8} > 0$ $\therefore \tan \frac{\pi}{8} = -1 + \sqrt{2}$	3 Marks Correct solution 2 Marks Deduce $\tan \frac{\pi}{8}$ is a root and finds the values of t 1 Mark Deduce $\tan \frac{\pi}{8}$ is a root of the equation